

DUL-92 — Stuck Pipe Historical Knowledge Pack

Synthetic historical offset-well record for RAG, similarity matching, and hazard explanation

WELL & EVENT SUMMARY

Field Metric	Value	Field Metric	Value
Well Name:	Duliajan-92 (DUL-92)	Field / Basin:	Duliajan Field, Upper Assam Shelf
Event Depth:	2,210 – 2,214.5 m MD	Formation Target:	Barail Group Shale
Primary Hazard:	Reactive shale / mechanical pipe entrapment	Rig Assigned:	OIL Rig 14 (2000 HP)
Spud Date:	12-Oct-2022	Target Depth:	3,520 m (MD)
Completion Date:	18-Jan-2023	Doc Type:	REPORT 1: QUICK SPOT KNOWLEDGE PACK

INCIDENT NARRATIVE

During a synthetic drilling scenario, the drill string became mechanically entrapped while pulling out of hole (POOH) at approximately 2,210 m in Barail Shale. Torque increased sharply and overpull was observed. The event is represented as a historical case that NWIS can retrieve when a current well enters a similar formation and shows a similar sensor signature.

RECORDED PARAMETERS AT INCIDENT TIME

Parameter	Historical Observed Value	Reference / Normal Baseline Context
Depth Interval	2,210 – 2,214.5 m MD	Event interval in Barail Shale
Mud Weight	1.16 g/cc (Water-Based)	1.24 g/cc EMW Synthetic Reference
Gamma Ray (GR)	118 API	Reference 45 – 60 API
Standpipe Pressure	2,850 psi	Reference ~2,100 psi (+750 psi Pack-off)
Rotational Torque	28.4 kN·m	Reference ~11.5 kN·m (+16.9 kN·m Entrapment Spike)
Smectite Clay Index	42% Smectite Content	Reference < 15% (Severe swelling hazard)

MITIGATION & OUTCOME RECIPE

Step 1: Spot 50 bbl Poly-Glycol / high-lubricity pill into annulus at 2,210 m depth.

Step 2: Soak for 4 hours while maintaining 10 tonnes tension and working jars in downward direction.

Step 3: Inhibited active mud system using KCl 8% and glycol 4% v/v treatment.

Step 4: Drill string freed successfully at approximately 09:15 hrs IST.

Step 5: Pump 30 bbl high-viscosity sweep and condition mud density to 1.25 g/cc.